

FUTURE — Roadmap to v1.0

Companion to: `x3d-first-time-setup-plan.md` (the baseline spec)

Created: 2026-07-25 (at plan v0.2)

Purpose: Sequenced milestone plan so work layers correctly — each version depends only on what came before it. Nothing here is a feature wishlist dump; the ordering is the point.

Layering rule: Content and tooling advance in lockstep. Every chapter written must have its verification tool built before the next chapter starts, otherwise we ship instructions we can't prove.

v0.3 – Truth Pass (asset intake + research)

Nothing gets written until we know what's actually true. This version produces zero user-facing content.

- [] Intake all supplied assets: detect script, X3D Undervolt Tester, iRacing kit scripts + README, troubleshooting notes/history, NVIDIA BSOD PSA, Backup-Interactive-v2.ps1
- [] Convert every [VERIFY] / [VALIDATED-DIRECTION] tag in the baseline to a confirmed claim or an explicit unknown
- [] Complete research items 1–8 (AGESA floor, MCR/Power Down pairing, readable-state audit, Zen 5 FCLK/UCLK consensus, C-states, HAPS, 9950X3D2 specs, per-SKU cache-CCD index)

- [] Resolve open decisions 5–12
- [] Audit detect script: what it already outputs vs. what the profile schema needs
- [] Lock repo name, create repo, license, contribution rules
- [] Establish claim-provenance convention (every factual line tagged: own-testing / vendor-doc / community-consensus)

Exit criteria: zero unresolved tags in the baseline spec. A written source for every claim we intend to publish.

v0.4 — Foundation Content (9950X3D only)

First user-readable output. Single chip, single path, deliberately narrow.

- [] Phase 0 BIOS chapter written end-to-end for 9950X3D
- [] BIOS Translation Table built for all 5 vendors (setting names + menu paths + BIOS-entry keys + CMOS-clear procedure per vendor)
- [] Memory training integrity chapter (the MCR/Power Down pillar) with the trained-value fingerprint capture procedure
- [] CO Path 1 (all-core conservative) written
- [] Boot integrity checklist finalized
- [] One-line BIOS explainer + access instructions

Exit criteria: a competent first-timer could follow it manually with no tools and reach a stable EXPO + conservative CO state.

v0.5 — Tooling Foundation + Dual-Format Engine

The verification layer that makes v0.4 provable. Build the output engine FIRST so every later tool inherits it.

- [] **Dual-format output engine** (shared module — Layer 1 human report, Layer 2 AI block with prompt header / state dump / explicit ask)
- [] History log schema + append-only writer (so "what's been tried" is always available to the AI block)
- [] Chip detect — extended from supplied script: class A/B/C mapping + profile lookup + board vendor detection (WMI baseboard)
- [] BIOS state auditor — reads live memory config, SoC voltage, whatever else v0.3 research proved readable

- [] Training fingerprint capture + compare (drift alarm across boots)
- [] Health report – runs the above, single pass/fail artifact

Exit criteria: every claim in v0.4 has a tool that confirms it. Every tool emits both formats.

v0.6 – Windows Baseline + Scheduling Verification

- [] Phase 1 Windows chapter: chipset driver stack / V-Cache Optimizer, Game Bar warning for Class B, Balanced power plan rationale, HAGS/VBS honest tradeoffs, NVIDIA driver hygiene (incl. the 0x61941 history)
- [] Scheduling verifier tool: V-Cache driver loaded, Game Mode state,

preferred-core ranking, test workload
lands on cache CCD

- [] Backup/restore integration (from Backup-Interactive-v2.ps1) – known-good state always recoverable before any change
- [] Handoff chapter → iRacing X3D Tuning kit, with the cross-links live in both repos

Exit criteria: full first-timer path complete for 9950X3D, BIOS through Windows, every step verified.

v0.7 – The Tuning Path (opt-in Path 2)

Only now, on a proven foundation, does deep tuning get exposed.

- [] Absorb/extend X3D Undervolt Tester into the suite

- [] Per-core guided stepping with per-CCD awareness (cache CCD vs frequency CCD ceilings)
- [] Recommendation engine: reads history log, suggests next offset step, flags asymmetries
- [] Testing protocol tiers implemented (Quick / Standard / Paranoid) with per-tier pass thresholds
- [] CoreCycler + idle-soak integration (the low-load CO failure mode)
- [] Every result routed through the dual-format engine

Exit criteria: a user can go from conservative to refined per-core tune without ever guessing a number unaided.

v0.8 — Chip Class Expansion

Framework was built for this since v0.1 — this version proves it by adding chips as

data, not code.

- [] Class A (single-CCD: 5800X3D, 7800X3D, 9800X3D) – profiles + content branching (scheduling chapters suppressed, not deleted)
- [] Class C (9950X3D2 dual-cache) – profiles + scheduling-behavior differences
- [] Remaining Class B (7900X3D, 7950X3D, 9900X3D)
- [] Zen 4 vs Zen 5 divergences documented where they matter
- [] Confirm adding a new SKU requires a profile entry only – no code changes. If it doesn't, refactor now.

Exit criteria: any current X3D owner is fully served; the next SKU AMD releases is a one-file addition.

v0.9 – Delivery, Polish, Beta

- [] Web delivery built (per decision #12
 - static vs. light JS for board-select/dropdown rendering)
- [] Board-select UX + "other terminology" dropdowns live
- [] README/first-impression pass (apply the readability lessons from the iRacing kit's Reddit-referred traffic)
- [] Beta group across at least 3 chip classes and 3 board vendors
- [] Failure-mode encyclopedia populated from beta reports
- [] Every tool re-run on clean installs to catch assumed-state bugs

Exit criteria: external users complete the guide without needing to ask us anything.

v1.0 – Release

- [] Full guide, all X3D classes, all 5 vendors, complete tool suite
 - [] Special-touch features from the v1.0 tier below shipped
 - [] Distribution: repo + web + release bundle
 - [] Announcement plan (subreddits, OpenMR, overclock.net Zen 5 X3D thread)
 - [] Maintenance policy documented (AGESA/driver table cadence, issue triage, how community submissions get reviewed)
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Special Touches

Ship in v1.0

1. The Panic Card (v0.4)

A printable one-page PDF: CMOS-clear procedure per vendor, BIOS-entry keys,

"your PC won't POST — do this" flow, and how to get back to last known-good.

Printed on purpose — the moment a user needs it most is the moment they can't open the website. Generated per-vendor once the board is selected. This is the single highest-trust item in the whole project; it says *we expect you to break it and we've got you*.

2. Resumable Progress State (v0.5)

BIOS tuning is a dozen reboots. The tooling tracks which steps are done, validated, or pending, and survives restarts — so the guide always opens at "here's where you left off, and here's what still needs validating." Doubles as the state source for the AI block.

3. Local LLM Mode (v0.5)

The AI troubleshooting block can pipe straight to a local model endpoint (LM Studio / Ollama-compatible) instead of

requiring copy-paste to a cloud service.
Fully offline tuning assistance, private by default. Fits the project's audience precisely and nothing else in this space does it.

4. Tune Passport (v0.7)

A single exportable card summarizing the validated tune: chip + class, board, memory config, trained fingerprint, per-CCD CO offsets, which test tiers passed and for how long. Shareable, and it's the honest answer to "what settings do you run?" — because it includes *what was actually validated*, not just what's typed in BIOS.

5. Regression Watchdog (v0.6)

Optional scheduled task: after any Windows update, driver install, or BIOS flash, re-run the fingerprint + scheduling verifier and alert on drift. This is the direct enforcement of the "guarantee the proper

tune at any given time" pillar — the tune stays proven, not just proven once.

6. Failure-Mode Encyclopedia (v0.9)

Symptom → likely cause → next test.

Seeded from our own history (the 0x61941 BSOD chain, iGPU confusion, overpowered-component diagnostics, wet-filament-style "everything looks fine but isn't" cases) and grown from beta reports. Each entry links to the exact tool that confirms or clears it, and each has a pre-built AI block.

7. Offline Bundle (v0.9)

A downloadable self-contained package (guide + tools + Panic Card). A fresh Windows install on a tuning bench often has no network yet, and telling someone to "go to the website" from a machine that won't boot is a design failure.

Post-1.0 stretch

8. Community Results Database

Opt-in anonymized submissions of Tune Passports → a real dataset of what each SKU actually tolerated, with sample sizes. Users get percentile feedback ("your CCD0 is in the top 30% of 9950X3D submissions we've seen"). Replaces forum anecdote with measured distribution. Requires v1.0's Passport format to be stable first, and needs a privacy/consent design done properly rather than bolted on.

9. Before/After Telemetry Capture

Tie into the iRacing kit to capture 1% lows and frametime consistency pre- and post-tune. Turns "feels smoother" into a number, which is the entire ethos of both projects. Deferred because it needs the iRacing kit's telemetry surface to be stable and cross-project versioning to be sorted.

10. Known-Good Versions Table

A separately maintained, dated table of AGESA / chipset driver / GPU driver combinations with known-good and known-bad flags. High ongoing maintenance cost — only commit to it if the maintenance policy in v1.0 can realistically sustain it. Better to ship nothing than a stale table people trust.

Anti-Goals (explicit non-features)

Recorded so scope creep gets refused by default:

- **No registry-hack soup.** Every change is documented, reversible, and justified with a measurement.
- **No chasing benchmark records.** Repeatability over peak numbers —

stated in the baseline principles,
enforced here.

- **No duplicating the iRacing kit.** Per-application tuning lives there. Link, never copy.
- **No unverifiable advice.** If we can't test it or source it, it doesn't ship — it stays a tagged unknown.
- **No blanket "disable this for performance."** Tradeoffs get stated honestly (security, features, stability) and the user decides.